

Drift-Sense — Evaluation on the Organizers' 25-Pair Set

AP-25 blind-style evaluation · 20 present / 5 absent · scored with the published Phase 2 rubric · **82.71 / 100** (before efficiency and docs)

1. Headline

The frozen submission (commit 33841c5, shipped checkpoint weights/driftsense.pt, sha256 e6506b7c...71ca8) was run through the organizers' required entry point `python register.py --input pairs.csv --output predictions.csv` on all 25 pairs, then scored by `judging/score_rubric.py` — the same implementation that produced the Phase 2 judging campaign's ranking table. No parameter was tuned against these labels.

Component	Weight	Earned	Share
Localisation (0.45·A + 0.55·B)	40	34.49	86.2%
Scale	10	10.00	100.0%
Rotation	10	8.89	88.9%
Rejection (F1 = 0.8889)	15	13.33	88.9%
Calibration (AUC = 1.0000)	10	10.00	100.0%
Subtotal	85	76.71	90.2%
Bonus — Set D ≥ 0.40 with A/B/F1/AUC ≥ 0.50	+6	6	granted
Bonus — rejection F1 ≥ 0.90	+4	0	missed
Total on the 100-pt scale	+10	82.71	82.71

Efficiency (5 pts) and the judged documentation block (10 pts) are scored outside this harness and are therefore excluded from the 82.71. On the campaign's own convention the 5-point efficiency component is earned by a median latency under the 5 s budget; this run measured a 1.89 s median on 4 threads (max 4.73 s), the same band as the 1.905 s that took full marks for the top-ranked team.

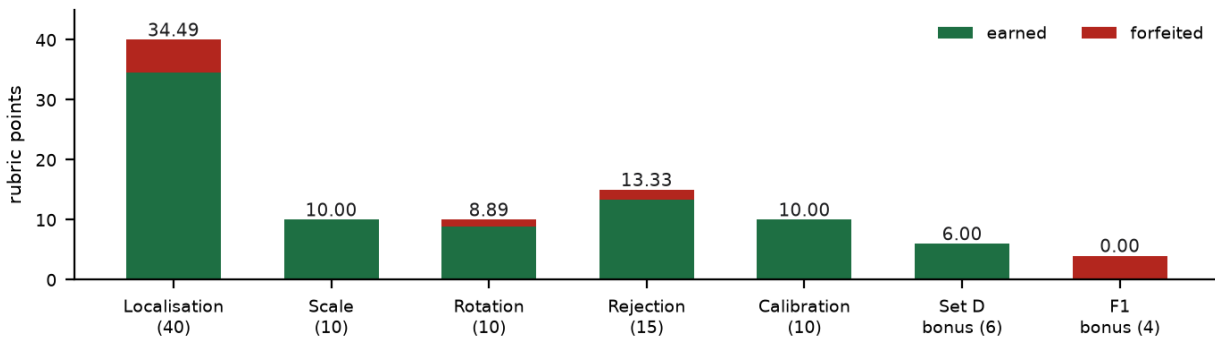


Figure 1 — Earned versus forfeited points, component by component. Localisation and the F1 bonus carry every forfeited point.

2. Context — against the organizers' own calibration

The set ships with a published calibration of the *naïve* ZNCC baseline. Rerunning that baseline here reproduces those published figures exactly (Set A 0.9778 vs 0.978 published; Set B 0.4444 vs 0.444; `core_credit` 0.6844 vs 0.684), which validates this harness against the organizers' own numbers before any comparison is drawn. The F1 column does not match (0.571 vs 0.833 published) and should not: the published value is computed over the parent 48-pair set, not this 25-pair cut.

Measure	Naive ZNCC baseline	Our submission	Δ
Set A localisation credit	0.9778	0.9111	-0.0667
Set B localisation credit	0.4444	0.8222	+0.3778
Localisation points /40	27.38	34.49	+7.11
Points on the 85-pt subtotal	59.56	76.71	+17.15
Rejection F1	0.5714	0.8889	+0.3175
Bonus points	0	6	+6
Total	59.56	82.71	+23.15

+23.15 points over the naive baseline, of which +6 is the Set D bonus gate and +4.76 is rejection F1. The learned model's advantage is concentrated exactly where the organizers said discrimination lives: on the degraded Set B the baseline's coarse grid clears the 0.55 gate on only 4 of 9 pairs, while the learned decode holds localisation credit at 0.8222 — and the baseline is *not* more accurate on those pairs; it simply declines them.

3. Where we are lagging

Ranked by points recoverable. Five distinct gaps, in priority order.

3.1 Localisation — the 1 px tier (≈ 8.5 pts at stake)

Localisation is the only component with meaningful mass left on the table: **34.49 of 40**. The cause is a single tier boundary. **9 of 20** present pairs land above 1.0 px and drop from credit 1.00 to 0.80 or 0.60. Only 11 pairs clear the 1 px tier; the naive baseline clears it 16 times on the same pairs. Mean error is 1.069 px against the baseline's 0.814 px. **On raw positional accuracy the naive baseline is currently ahead of us on Set A** (0.9778 vs 0.9111), and that is the single most actionable finding in this report.

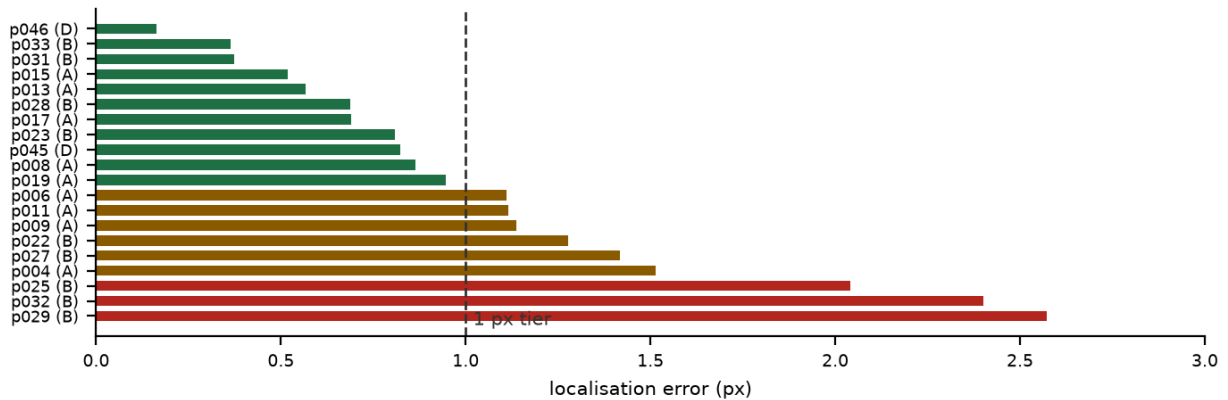


Figure 2 — Per-pair localisation error, present pairs. Green ≤ 1 px (full credit), amber ≤ 2 px (0.80), red > 2 px.

Pair	Set	Architecture	Sev	err (px)	credit	x err	y err	score
p029	B	finfet_22nm	2	2.572	0.60	-2.501	+0.598	0.731
p032	B	dram_1x	4	2.401	0.60	+2.349	+0.497	0.495
p025	B	dram_legacy	3	2.040	0.60	+1.932	+0.654	0.729
p004	A	dram_loose	1	1.513	0.80	+1.415	+0.534	0.903
p027	B	finfet_7nm	3	1.416	0.80	+1.314	+0.528	0.646
p022	B	dram_loose	3	1.277	0.80	+1.188	+0.470	0.624
p009	A	finfet_10nm	1	1.137	0.80	+0.961	+0.607	0.845
p011	A	finfet_14nm	1	1.116	0.80	+0.995	+0.506	0.854

Pair	Set	Architecture	Sev	err (px)	credit	x err	y err	score
p006	A	dram_compact	1	1.110	0.80	+0.978	+0.524	0.890

3.2 A systematic x-axis bias in the sub-pixel placement

The error is not isotropic. Signed error averages **dx = +0.610 px** (std 0.984) and **dy = +0.486 px** (std 0.159); dx is positive on 18 of 20 pairs and dy on 20 of 20, with a tight dy distribution ($\sigma = 0.16$ px) that is a fingerprint of a constant convention offset rather than per-pair noise. The organizers' own baseline shows no such offset (dx = +0.112, dy = -0.086 on the same pairs). On the narrow pairs the bias is the *entire* deficit: p009 sits at 1.137 px of which 0.961 px is x, p011 at 1.116 px of which 0.995 px is x — halving the x offset alone would move both into the 1 px tier.

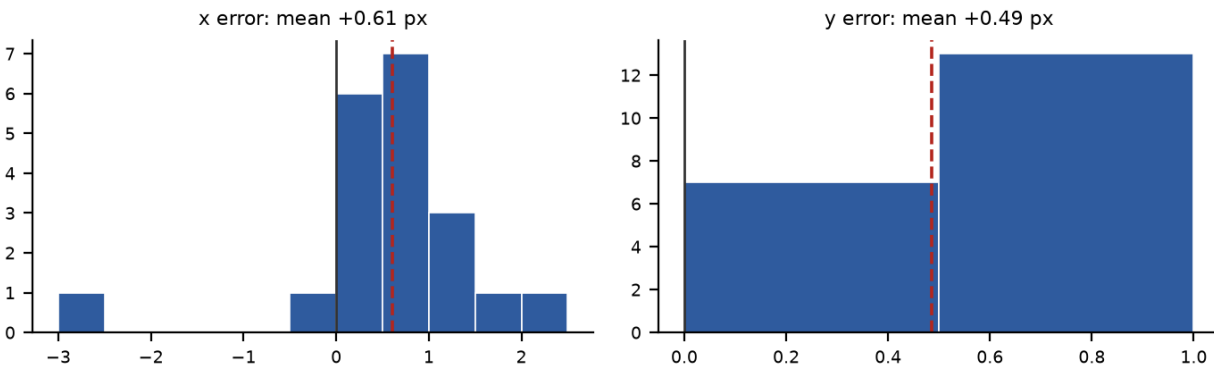


Figure 3 — Signed error distributions. A zero-centred cloud is correct; ours is shifted on both axes, the baseline's is not.

This is stated as a diagnostic, not a free fix: on the blind set no ground truth exists, so a constant cannot simply be subtracted. It localises the defect to the sub-pixel placement stage, and it is checkable against held-out data.

3.3 The shipped row-drift refinement fails to reproduce on this set

SHIPPED_SUBPIXEL_ROWS = True re-replaces x on the scan row the label is defined against, and its promotion record cites a full 2,500-pair paired A/B worth +0.589 localisation points with 95% CI [+0.410, +0.773]. A paired A/B on these 25 pairs **goes the other way**: it moves x on 10 of 20 pairs while y never moves, and it hurts more than it helps.

Arm	Set A credit	Set B credit	Loc. points	Set A mean err	Set B mean err
rows ON (shipped)	0.9111	0.8222	34.49	0.940	1.326
rows OFF	0.9111	0.8667	35.47	0.956	1.170

On this set the refinement is worth **-0.98 localisation points**. It hurts 4 pairs, helps 5, and is a no-op on 11; its largest single harm is p029, which it moves 2.90 px in x and 0.72 → 2.57 px of error. Its largest help is p046 (0.72 → 0.16 px), though that pair is Set D and therefore bonus-only. The mechanism is visible in the promotion evidence itself: the drift row is recovered from the *search* trace, so the correction is a property of each generator's raster-drift model, and these 25 pairs come from a different generator build than the 2,500-pair campaign. This is a **generalisation gap, not a bug** — and it is worth up to +1.0 point.

3.4 The rejection F1 ≥ 0.90 bonus (4 pts) is one pair away

Rejection F1 is **0.8889** (tp/fp/fn = 4/0/1) against the 0.90 gate, so the +4 bonus is forfeited by a single false negative at p035. p035 is a severity-0 *dram_1x* absent pair — a clean, high-dose decoy whose absent-family signature is the hardest to separate — and it scored **0.2208** against the 0.18 gate, i.e. only 0.04 above the decision line. Correctly declining it alone would take F1 to 1.0000 and restore the full +4, moving the total from 82.71 to 86.71.

Pair	Set	Architecture	Sev	absent?	score	0.18 gate	verdict
p035	C	dram_1x	0	yes	0.2208	0.1800	ACCEPTED ✗
p036	C	dram_dense	2	yes	0.1675	0.1800	declined ✓

Pair	Set	Architecture	Sev	absent?	score	0.18 gate	verdict
p038	C	dram_legacy	0	yes	0.0822	0.1800	declined ✓
p037	C	dram_wide	1	yes	0.0000	0.1800	declined ✓
p039	C	dram_compact	2	yes	0.0000	0.1800	declined ✓

Note the margin structure: the four correctly-declined absent pairs scored 0.000, 0.000, 0.082 and 0.168, while the lowest-scoring *present* pair scored 0.387. The separation is real and the gate sits in a genuinely empty band — p035 is an outlier, not a threshold problem. Moving the gate up to catch it would cost a true positive at 0.387 and lose more than the 4 points it recovers.

3.5 Rotation is the weaker pose axis

Scale is fully solved: all 20 credited pairs land inside the 1% tier, median 0.199%. Rotation is not: **6 of 20** pairs miss the 0.25° tier (median 0.155°, worst 0.458°), costing 1.11 points. The failing pairs are p013, p023, p027, p028, p029 and p046 — five of the six are severity 3–4. There is no scale work left to do; every remaining pose point is a rotation-accuracy problem.

4. Stratified view

Where the loss concentrates, by severity, architecture family and pose.

By severity

severity	n	mean err (px)	≤ 1 px	loc. credit
0	2	0.843	2/2	1.0000
1	9	0.862	5/9	0.9111
2	2	1.472	1/2	0.8000
3	3	1.578	0/3	0.7333
4	4	1.065	3/4	0.9000

By architecture family

family	n	mean err (px)	≤ 1 px	loc. credit
DRAM	10	1.213	5/10	0.8600
FinFET	10	0.926	6/10	0.9000

By set

set	n	mean err (px)	≤ 1 px	loc. credit
A	9	0.940	5/9	0.9111
B	9	1.326	4/9	0.8222
D	2	0.493	2/2	1.0000

Two readings worth recording. First, **severity is not the driver**: level 4 pairs average a *lower* error (1.065 px, 3 of 4 inside 1 px) than level 2 (1.472 px) or level 3 (1.578 px, 0 of 3 inside 1 px). The organizers predicted this — severity compresses peak *confidence* while peak *position* holds — and our numbers reproduce it. The middle of the severity range is where we actually lose, which is the opposite of where tuning effort would naturally go. Second, DRAM is slightly worse than FinFET on mean error (1.213 vs 0.926 px) but its lower credit is almost entirely the two outliers p025 and p032; the family gap is not structural at n = 10.

5. Full per-pair record

Pair	Set	Arch	Sev	pres	found	err	credit	x err	y err	scale %	rot°	score
p004	A	dram_loose	1	1	1	1.513	0.80	+1.415	+0.534	0.383	0.118	0.9028
p006	A	dram_compact	1	1	1	1.110	0.80	+0.978	+0.524	0.030	0.019	0.8902
p008	A	finfet_10nm	0	1	1	0.863	1.00	+0.699	+0.506	0.395	0.153	0.8746
p009	A	finfet_10nm	1	1	1	1.137	0.80	+0.961	+0.607	0.000	0.188	0.8450
p011	A	finfet_14nm	1	1	1	1.116	0.80	+0.995	+0.506	0.000	0.127	0.8537
p013	A	finfet_28nm	1	1	1	0.568	1.00	+0.194	+0.533	0.044	0.348	0.7900
p015	A	dram_dense	1	1	1	0.520	1.00	+0.226	+0.468	0.057	0.038	0.9175
p017	A	dram_wide	1	1	1	0.690	1.00	+0.429	+0.540	0.077	0.220	0.9012
p019	A	dram_compact	1	1	1	0.945	1.00	+0.730	+0.601	0.220	0.114	0.8771
p022	B	dram_loose	3	1	1	1.277	0.80	+1.188	+0.470	0.002	0.071	0.6237
p023	B	dram_wide	4	1	1	0.807	1.00	+0.570	+0.572	0.526	0.355	0.5590
p025	B	dram_legacy	3	1	1	2.040	0.60	+1.932	+0.654	0.214	0.180	0.7288

Pair	Set	Arch	Sev	pres	found	err	credit	x err	y err	scale %	rot°	score
p027	B	finfet_7nm	3	1	1	1.416	0.80	+1.314	+0.528	0.222	0.344	0.6458
p028	B	finfet_14nm	4	1	1	0.686	1.00	-0.484	+0.486	0.320	0.416	0.5003
p029	B	finfet_22nm	2	1	1	2.572	0.60	-2.501	+0.598	0.016	0.458	0.7311
p031	B	finfet_45nm	2	1	1	0.373	1.00	+0.199	+0.316	0.185	0.100	0.7649
p032	B	dram_1x	4	1	1	2.401	0.60	+2.349	+0.497	0.652	0.156	0.4950
p033	B	finfet_10nm	4	1	1	0.364	1.00	+0.363	+0.028	0.231	0.080	0.3867
p035	C	dram_1x	0	0	1	—	0.00	—	—	—	—	0.2208
p036	C	dram_dense	2	0	0	—	0.00	—	—	—	—	0.1675
p037	C	dram_wide	1	0	0	—	0.00	—	—	—	—	0.0000
p038	C	dram_legacy	0	0	0	—	0.00	—	—	—	—	0.0822
p039	C	dram_compact	2	0	0	—	0.00	—	—	—	—	0.0000
p045	D	dram_1x	0	1	1	0.823	1.00	+0.535	+0.625	0.062	0.103	0.8817
p046	D	finfet_10nm	1	1	1	0.163	1.00	+0.111	+0.120	0.854	0.345	0.8643

Reproduce. First `python register.py --input evaluation/ap25/pairs.csv --output predictions.csv`, then `python judging/score_rubric.py --pred predictions.csv --gt ground_truth.csv --manifest manifest_jury.csv`.

Inputs, both decoder arms and the naive baseline are frozen under `evaluation/ap25/`.